

Greedy Solution to
Kanpsack Problem

Knapsack Problem

- Given a set of items I with corresponding profit p_i and weight w_i for every item i and knapsack capacity C . Find out I' , a subset of I , which gives maximum profit and total weight is not more than capacity C .
- There are 2 versions of the problem:
 1. Fractional Knapsack where items are divisible and you can take any fraction of the item as part of your solution.
 2. 0/1 Knapsack where items are indivisible and only either it can be taken fully or not in the solution.

Fractional Knapsack Problem

- Formally, the problem can be stated as

$$\text{maximize } \sum_{1 \leq i \leq n} p_i x_i$$

$$\text{subject to } \sum_{1 \leq i \leq n} w_i x_i \leq C$$

$$\text{and } 0 \leq x_i \leq 1, 1 \leq i \leq n$$

- The profits and weights are positive numbers

Example 1

Item	i_1	i_2	i_3
Profit	25	24	15
Weight	18	15	10

Sack Capacity (C) = 20

Example 1 – Solution 1

- **Select** items in the non-increasing order of profit

Iteration	Item added	Profit	Remaining Capacity
0	$\{\}$	0	20
1	$\{i_1\}$	25	2
2	$\{i_1, (2/15)*i_2\}$	$25 + 3.2 = 28.2$	0

- Total Profit = **28.2**

Example 1 – Solution 2

- **Select** items in the non-decreasing order of weight

Iteration	Item added	Profit	Remaining Capacity
0	$\{\}$	0	20
1	$\{i_3\}$	15	10
2	$\{i_3, (2/3)*i_2\}$	$15 + 16 = 31$	0

- Total Profit = **31**

Example 1 – Solution 3

- **Select** items in the non-increasing order of **profit / weight** ratio

Item	i_1	i_2	i_3
Profit	25	24	15
Weight	18	15	10
Profit / Weight	1.39	1.6	1.5

Example 1 – Solution 3

- **Select** items in the non-increasing order of **profit / weight** ratio

Iteration	Item added	Profit	Remaining Capacity
0	$\{\}$	0	20
1	$\{i_2\}$	24	5
2	$\{i_2, (1/2)*i_3\}$	$24 + 7.5 = 31.5$	0

- Total Profit = **31.5**

Example 2

Item	i_1	i_2	i_3	i_4
Profit	10	6	8	12
Weight	5	2	3	5
Profit / Weight	2	3	2.66	2.4

Sack Capacity (C) = 5

Greedy Algorithm for Fractional Knapsack Problem

Data: $I[1:n]$ item array, $p[1:n]$ profit, $w[1:n]$ weight, C Knapsack Capacity, $s[1:n]$ solution vector

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1  $I[1:n] = \text{Sort}(I[1:n], p[1:n], w[1:n])$  ;  $\longrightarrow$   $O(n \log n)$ 
2 for  $i \leftarrow 1$  to  $n$  do
3    $s[I[i]] = 0$  ;  $\longrightarrow$   $O(n)$ 
4 for  $i \leftarrow 1$  to  $n$  do
5   if  $w[I[i]] \leq C$  then
6      $s[I[i]] = 1$  ;
7      $C = C - w[I[i]]$  ;
8      $P = P + p[I[i]]$  ;  $\longrightarrow$   $O(n)$ 
9   else
10     $s[I[i]] = C/w[I[i]]$  ;
11     $C = 0$  ;
12     $P = P + s[I[i]] * p[I[i]]$  ;
13 Print Total Profit  $P$  ;  $\longrightarrow$   $O(1)$ 
14 return  $s$  ;
```

$O(n \log n)$

References

1. A Book – “**Fundamentals of Computer Algorithms**” by Horowitz, Sahni, Rajasekaran